

# Chihway Chang

---

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                    |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LAST UPDATED              | September 7, 2026                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                    |
| CONTACT INFORMATION       | University of Chicago<br>William Eckhardt Research Center 451<br>5640 South Ellis Avenue<br>Chicago, IL 60637<br>USA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | DOB: 1985/03/12<br>+1-773-702-3369<br><a href="mailto:chihway@uchicago.edu">chihway@uchicago.edu</a><br><a href="https://chihway.github.io">https://chihway.github.io</a><br><a href="http://surveys.uchicago.edu">http://surveys.uchicago.edu</a> |
| ACADEMIC APPOINTMENTS     | Co-PI and Deputy Director for Nurturing Next Generation, The NSF-Simons AI Institute for the Sky (SkAI Institute) 2025 – present<br>Associate Professor, Department of Astronomy and Astrophysics (A&A), University of Chicago 2025 – present<br>Assistant Professor, Department of Astronomy and Astrophysics (A&A), University of Chicago 2018 – 2025<br>Senior Member, Kavli Institute of Cosmological Physics (KICP), University of Chicago 2018 – present<br>Clare Boothe Luce Assistant Professor, Department of Astronomy and Astrophysics (A&A), University of Chicago 2018 – 2023<br>KICP Fellow, KICP, University of Chicago 2016 – 2018<br>Postdoctoral Fellow, Institute of Astronomy, ETH Zurich 2013 – 2016                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                    |
| EDUCATION                 | Physics Department, Stanford University Ph. D. 2013<br>Thesis: <i>Systematic Effects in Weak Lensing Measurements for Future Optical Surveys</i><br>Advisors: Steven M. Kahn & Rafe H. Schindler<br><br>Physics Department, National Taiwan University B. S. 2007                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                    |
| HONOURS AND RECOGNITIONS  | Scialog Fellow (Early Science with the LSST) 2024<br>Editor's choice in Physical Review Letters 2023<br>DOE Early Career Award 2021<br>Clare Boothe Luce Assistant Professor 2018<br>KICP Fellow 2016<br>Editor's choice in Physical Review Letters 2015<br>62nd Meeting of Nobel Laureate, Lindau, Germany 2012<br>Presidential Awards, National Taiwan University 2004 – 2007                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                    |
| SCIENTIFIC COLLABORATIONS | <p>I am a core member and recognized builder (i.e., foundational contributor) in two large scientific collaborations – the Dark Energy Survey (DES) and the Rubin Observatory Legacy Survey of Space and Time Dark Energy Science Collaboration (LSST DESC). I am also a member of the Roman Space Telescope Project Infrastructure Team (PIT) team “Maximizing Cosmological Science with the Roman High Latitude Imaging Survey”.</p> <p><b>Dark Energy Survey (DES):</b> I have been part of DES since 2013 and have been active in the weak gravitational lensing working group, as well as the general area of cosmology from large-scale structure. I am one of the two Science Committee Co-Chairs in DES, which directs and coordinates all scientific efforts within DES.</p><br>DELVE (DECam Local Volume Exploration Survey) Builder 2023 – present<br>DES Science Committee Co-Chair 2021 – present<br>DES Builder 2019 – present<br>DES Membership Committee 2019 – 2021<br>DES Management Committee 2018 – 2021 |                                                                                                                                                                                                                                                    |

DES Weak Lensing Mass Mapping Analysis Team Lead

2016 – 2020

**Rubin Observatory Legacy Survey of Space and Time Dark Energy Science Collaboration (LSST DESC):** I have been active in LSST DESC since 2016 (and active in the LSST science working groups before the start of DESC in 2010). I primarily work in the weak gravitational lensing working group, as well as the general area of cosmology from large-scale structure. Initiated through my term as the Deputy Analysis Coordinator 2021-2023, I co-lead the SStatic Analysis Roundtable (STAR) group.

|                                                                             |                |
|-----------------------------------------------------------------------------|----------------|
| LSST DESC Builder                                                           | 2024 – present |
| DESC Speaker Bureau                                                         | 2024 – present |
| Rubin in-kind contributor team (Science validation of PSF characterization) |                |
| <a href="https://sitcomtn-050.lsst.io/">https://sitcomtn-050.lsst.io/</a>   | 2024 – present |
| LSST DESC Deputy Analysis Coordinator                                       | 2021 – 2023    |
| LSST DESC Collaboration Council                                             | 2020 – 2021    |
| LSST DESC Weak Lensing Working Group Coordinator                            | 2018 – 2020    |

|                                                                   |                                                                                                                               |                |
|-------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|----------------|
| PROFESSIONAL<br>SERVICE AND<br>RECENT<br>CONFERENCES<br>ORGANIZED | Cosmological Synergies from Multiple Overlapping Probes Observed by Next Generation Surveys LOC/SOC                           | 2025           |
|                                                                   | Roman PIT meeting SOC                                                                                                         | 2024           |
|                                                                   | Fundamental Physics meets Current and Future Facilities in Cosmology (Brazil) SOC                                             | 2024           |
|                                                                   | KICP-20 “Cosmology past, present, and future” (KICP) LOC/SOC                                                                  | 2024           |
|                                                                   | UChicago-Taiwan student research program in the Sciences (UCTS) organizer                                                     | 2024           |
|                                                                   | Spec-S5 Instrumentation Workshop (KICP) LOC                                                                                   | 2024           |
|                                                                   | Lensing at Different Scales Workshop (KICP) SOC                                                                               | 2023           |
|                                                                   | DES Y6KP Workshop (KICP) LOC                                                                                                  | 2023           |
|                                                                   | LSS × CMB Workshop (Kyoto) SOC                                                                                                | 2023           |
|                                                                   | DES Collaboration Meeting (Portsmouth) SOC                                                                                    | 2023           |
|                                                                   | LSST DESC Collaboration Meeting (KICP) LOC Co-Chair/SOC                                                                       | 2022           |
|                                                                   | CMB-S4 Collaboration Meeting (remote) SOC                                                                                     | 2020           |
|                                                                   | Snowmass 2021 Topical Group Convener (CF6: Dark Energy and Cosmic Acceleration: Complementarity of Probes and New Facilities) | 2020 – 2021    |
|                                                                   | Cosmic Controversy (KICP) SOC                                                                                                 | 2019           |
|                                                                   | Joint SPT-DES Analysis Workshop (KICP)                                                                                        | 2018           |
|                                                                   | DES Collaboration Meeting (Chicago) LOC                                                                                       | 2017           |
|                                                                   | Review panel for NSF, DOE                                                                                                     | 2020 – present |
|                                                                   | Reviewer for MNRAS, ApJ, Nature Astronomy, DES/DESC Collaboration                                                             | 2014 – present |

|                                    |                                                                                                                                                                             |             |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| SCIENTIFIC<br>PROPOSALS<br>AWARDED | Research grants that I have led or contributed to significantly. Total funding through grants, awards, and scholarships: <b>\$3.0M</b> .                                    |             |
|                                    | Roman Cycle I: A New Look at Dark Matter in the Roman Era: Joint Inference of High-Resolution Mass Map and Cosmology\$150k [PI]                                             | 2027 – 2029 |
|                                    | Roman Cycle I: Dynamical state of galaxy clusters in the Roman COSMOS Deep Field \$150k [Co-PI]                                                                             | 2027 – 2030 |
|                                    | DOE FY2025 HEP Comparative Review, Theoretical High Energy Physics Research at the University of Chicagos, \$330k [Co-PI]                                                   | 2025 – 2028 |
|                                    | France and Chicago Collaborating in the Sciences (FACCTS) grant: Breaking Through Blending: A Multi-Survey Approach to Dark Energy with LSST, Euclid, and Roman, \$12k [PI] | 2025 – 2027 |

NSF AI in Astronomy Institute 2024, AI for the Sky: The SkAI Institute, \$20M for the entire institute (not counted above) [Senior Personnel] 2024 – 2029

NSF AAG 2024, A Novel Approach to Constraining Baryonic Feedback: Coherent Field-Level Inference of Lensing and Electron Pressure, \$458k [PI] 2024 – 2027

NASA Roman Space Telescope Project Infrastructure Team (PIT) 2023, Maximizing Cosmological Science with the Roman High Latitude Imaging Survey, \$323k [Co-I] 2024 – 2029

NSF AAG 2022, Collaborative Research: Combining Galaxy and Cosmic Microwave Background Surveys for Precise and Robust Constraints on Cosmology, \$250k [PI] 2023 – 2026

France and Chicago Collaborating in the Sciences (FACCTS) grant: Large-Scale Structure Simulations for Next Generation Cosmological Modeling, \$25k [PI] 2022 – 2023

DOE Early Career Award 2021, Towards Robust Cosmology from Large-Scale Structure with Galaxy Surveys, \$750k [PI] 2021 – 2026

NSF AAG 2020, Collaborative Research: Cosmic Shear on Extremely Large Scales with the Dark Energy Camera, \$360k [PI] 2021 – 2024

DOE FY2020 HEP Comparative Review, Cosmology with Galaxy and CMB Surveys: Mitigating Systematic Effects Through Combining Datasets, \$125k [PI] 2020 – 2022

University of Chicago Center for Data and Computing (CDAC) Data Science Discovery Grant – When Technology Transforms Society: Considering the Societal and Ethical Impacts of Quantum Computing and AI, \$15k [PI] 2019 – 2020

UChicago seed funding, Mapping Dark Matter with the Faintest Galaxy, \$40k [PI] 2018 – 2019

## TEACHING

The SkAI Institute Satellite Early Education for Discovery (SEED) Program: *Lecturer* Summer 2026

ASTR 298 (UChicago), Research Seminar: *Guest Lecturer* Spring 2024

ASTR 340 (UChicago) Statistical Methods in Astrophysics: *Instructor* Winter 2024, 2025

ASTR 18850 (UChicago) Interpreting Nature – on the Relation between Art and Science: *Instructor* Spring 2023, 2024, 2025, 2026 (Paris)

ASTR 406 (UChicago) Gravitational Lensing: *Instructor* Fall 2020

ASTR 133 (UChicago) Introduction to Astrophysics: *Instructor* Spring 2019, 2020, 2021, 2022

PHYS 372 (UChicago), Space Physics and Astrophysics: *Guest Lecturer* Fall 2016, 2018

Astrophysics I (ETH): *Substitute Lecturer* Fall 2015, 2014

Astrowoche (ETH): *Teaching Assistant* Spring 2016, 2015

Cosmological Probes (ETH): *Teaching Assistant and Substitute Lecturer* Spring 2014

Physics 21 (Stanford), Mechanics and Heat: *Teaching Assistant* Fall 2011

Physics 23 (Stanford), Electricity and Optics: *Teaching Assistant* Winter 2007

Physics 41 (Stanford), Light and Heat: *Teaching Assistant* Fall 2007

## ADVISING AND MENTORING

### Postdocs and research scientists:

Chun-Hao To September 2024 – present

Eric & Wendy Schmidt AI Fellow at UChicago (serve as primary mentor). Primarily working in DES and Roman on simulations and cluster cosmology. Expert in combining galaxy cluster datasets across different wavelengths.

Marco Gatti August 2024 – October 2025  
KICP Fellow (serve as primary mentor). Primarily working in DES on weak lensing. Expert in higher-order statistics and new statistical techniques to extract cosmological information from data. Will also be transferring knowledge to LSST DESC.

Giulia Giannini October 2022 – October 2025  
Postdoc primarily working in DES on weak lensing and large-scale structure cosmology. Expert in redshift calibration and galaxy-galaxy lensing. Transferring knowledge to LSST DESC.

Yuuki Omori January 2021 – present  
Working with SPT and my group. Expert in CMB lensing and cross-correlation between galaxy and CMB datasets. Also working on field-level inference for weak lensing. Transitioned from postdoc to research scientist in 2024.

Lucas Secco September 2020 – January 2024  
KICP Fellow (served as primary mentor). Worked on DES cosmoc shear analysis and then helped with the DELVE cosmic shear analysis effort. Investigated lensing higher-order statistics as well as cosmic tensions. Moved on to Boston Consulting Group.

Judit Prat October 2019 – December 2023  
Postdoc working on both DES and LSST DESC. Expert in weak lensing and large-scale structure cosmology and contributed to major infrastructure in both DES and DESC pipelines. Moved on to a Eric & Wendy Schmidt AI Fellow at UChicago (served as primary mentor), then a Nordita Fellow.

#### **Graduate students:**

Shrihan Agarwal December 2023 – present  
PhD, UChicago A&A. Studying the impact of spatially varying systematic effects in weak lensing data for DES, DELVE and LSST. Won an 2023 NSF Graduate Research Fellowship.

Johnny Pitocco August 2023 – present  
PhD, UChicago A&A. Joint advised with Jeff McMahon working on cosmology analysis with DES Y3 data and ACT DR6 lensing.

Jazmine Jefferson June 2022 – August 2026  
PhD, UChicago A&A. Working on reanalysis of Stage-III weak lensing data sets in the context of LSST. Won an 2021 NSF Graduate Research Fellowship.

Dhayaa Anbajagane September 2020 – August 2026  
PhD, UChicago A&A. Working on a wide array of topics in DES and DELVE: cluster profiles, higher-order statistics in lensing, and connection with early universe physics. Also leading the DELVE cosmic shear analysis. Won an 2020 NSF Graduate Research Fellowship and a 2023 CCAPP Price Prize in Cosmology and Astrophysics.

Georgios Zacharegkas October 2018 – August 2023  
PhD, UChicago A&A. Studied the modeling of small-scale information in galaxy-galaxy lensing and galaxy clustering through the Halo Occupation Distribution framework and constrained the models with DES data. Moved on to a postdoc position at the Argonne National Lab.

Dimitrios Tanoglidis October 2017 – October 2019  
PhD, UChicago A&A (primary advisor Alex Drlica-Wagner). Worked on investigating sample selection in clustering measurements. Moved on to Data Science Fellow at UPenn.

Yi Zhao September 2018 – December 2018  
Master, UChicago PSD. Worked on galaxy-galaxy lensing measurement in simulations.

Marco Gatti October 2017 – May 2019  
PhD, Barcelona A&A (primary advisor Ramon Miquel). Using DES data to perform higher-order statistics measurement and modeling. Moved on to a postdoc position at UPenn.

Claudio Bruderer October 2016 – June 2018  
PhD, ETH Zurich A&A (primary advisor Alexandre Refregier). Develop a novel weak lensing shear estimation method based-on fast forward-modeling.

**Undergraduate and post-bacc students:**

Harjas Sandhu March 2024 – March 2025  
Major in Physics, UChicago. Joint advise with Eric Baxter. Working on using machine learning to perform background estimation for CMB datasets.

Kate Overdeck June 2023 – December 2023  
Major in A&A, UChicago. Joint advised with Giulia Giannini and Josh Frieman to study galaxy clustering split by colors.

Louise Gagnon June 2022 – September 2023  
Major in A&A, UChicago. Joint advised with Judit Prat on combining gravitational wave data and galaxy surveys.

Jon Shao June 2022 – June 2023  
Major in Physics, UChicago. Joint advised with Dhayaa Anbajagane on looking at how baryonic effects impact cluster characteristics. Moved on to PhD program at Caltech.

Kihana Wilson December 2021 – June 2023  
Major in A&A, UChicago. Joint advised with Judit Prat on studying the impact of deflection on weak lensing and clustering measurements.

Raul Basilides Gomez Del Estal Teixeira July 2021 – June 2024  
Major in A&A, UChicago. Working on redshift estimation and calibration for the DELVE cosmic shear project. Moved on to PhD program at Duke.

Nathalie Chicoine July 2020 – August 2024  
Major in A&A, UChicago. Joint advised with Judit Prat on measuring lensing around low surface brightness galaxies (LSBG), as well as cosmology from DELVE. Moved on to PhD program at UPitt.

Zhuoqi Zhang December 2019 – June 2023  
Major in Physics, UChicago. Worked on combining CMB lensing and galaxy surveys for LSST DESC. Also worked on DELVE cosmic shear testing. Moved on to PhD program at Stanford.

Ariel Amsellem January 2019 – December 2020  
Major in Physics, UChicago. Studied the splashback radius in simulations and compared to data. Moved on to PhD program at CMU.

Benjamin Levine December 2019 – 2021  
Major in A&A, UChicago. Studies the effect of blending on cosmology from galaxy clustering in LSST. Moved on to PhD program at Stonybrooks.

Rebecca Chen October 2017 – May 2019  
Major in A&A, UChicago. Worked on PSF modeling and measuring moments in weak lensing mass maps. Moved on to PhD program at Duke.

|                 |                                                                                                                                                                                                              |                   |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| PUBLIC OUTREACH | Picture an Astronomer opening panel                                                                                                                                                                          | 2025              |
|                 | Adler SVL volunteer                                                                                                                                                                                          | 2024 – present    |
|                 | UChicago myCHOICE workshop: Business of Running a Research Group                                                                                                                                             | 2023              |
|                 | Lecture and volunteer at Space Explorer                                                                                                                                                                      | 2023 – present    |
|                 | PSD Adler panel discussion                                                                                                                                                                                   | 2023              |
|                 | Interview with the Chicago Council on Science and Technology (C2ST)                                                                                                                                          | 2022              |
|                 | Talk at the Ryerson Astronomical Society at UChicago                                                                                                                                                         | 2021              |
|                 | Talk at Project Exploration (middle school students)                                                                                                                                                         | 2021              |
|                 | Kavli Community Forum career panel                                                                                                                                                                           | 2021              |
|                 | GRADUCon 2021 panel                                                                                                                                                                                          | 2021              |
|                 | UChicago PSD Women in STEM panel                                                                                                                                                                             | 2021              |
|                 | KIPAC career panel                                                                                                                                                                                           | 2020              |
|                 | Panel at SLAC public lecture series                                                                                                                                                                          | 2020              |
|                 | Career panel in DES, SLAC                                                                                                                                                                                    | 2018, 2020        |
|                 | Co-organizer of workshop “When Technology Transforms Society: Considering the Societal and Ethical Impacts of Quantum Computing and AI”                                                                      | 2019              |
|                 | APS Conferences for Undergraduate Women in Physics (CUWiP) Panel                                                                                                                                             | 2019              |
|                 | Public talk at the <i>Art of Science</i> series                                                                                                                                                              | 2018              |
|                 | Volunteer at the Adler Planetarium: <i>Astronomy Conversations</i>                                                                                                                                           | 2016 – 2020       |
|                 | DES outreach program <i>DarkBites: Lead illustrator</i> , 50+ illustrations<br>( <a href="https://www.darkenergysurvey.org/education/darkbites/">https://www.darkenergysurvey.org/education/darkbites/</a> ) | 2014 – 2015, 2021 |
|                 | SLAC tour guide                                                                                                                                                                                              | 2010 – 2013       |

|       |                                                                                                                |
|-------|----------------------------------------------------------------------------------------------------------------|
| PRESS | Scientists release newly accurate map of all the matter in the universe [UChicago, 2023]                       |
|       | Less clumpy universe may suggest existence of mysterious forces [The Guardian, 2023]                           |
|       | Cosmological Parameters Improved by Combining Data [APS, 2023]                                                 |
|       | Connecting the dots in the sky could shed new light on dark matter [SLAC, 2020]                                |
|       | Dark Energy Survey reveals most accurate measurement of dark matter structure in the universe [Fermilab, 2017] |
|       | New map of Universe’s dark matter [BBC, 2017]                                                                  |
|       | Scientists Unveil New Inventory of Universe’s Dark Contents [Quanta, 2017]                                     |
|       | Erhellendes zur Dunklen Materie[Spiegel, 2017]                                                                 |
|       | The Halo Boundary of Galaxy Clusters in SDSS [KICP, 2017]                                                      |
|       | Dark matter map unveils first results [BBC, 2015]                                                              |

|                |                                                                                                                                            |                    |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| SELECTED TALKS | AAS DES session, Pheonix, AZ                                                                                                               | January 6, 2026    |
|                | Astronomy Seminar, Purdue University, IN                                                                                                   | November 10, 2025  |
|                | Invited talk at "COSMO-25: 28th International Conference on Particle Physics & Cosmology", Pittsburgh, PA                                  | October 13, 2025   |
|                | Invited talk at "Beyond-2pt Statistics Meet Survey Systematics", Tokyo, Japan                                                              | September 16, 2025 |
|                | Invited talk at "Fundamental Physics meets Current and Future Facilities in Cosmology", Sao Paolo, Brazil                                  | December 2, 2024   |
|                | Lecture at the UChicago Data Science Institute Summer School, Chicago, IL                                                                  | July 18, 2024      |
|                | Invited talk at Gordon Research Conference "Imaging and Visualization at the Junction of Physics, Engineering and Data Science", Newry, MN | June 9, 2024       |

|                                                                         |                    |
|-------------------------------------------------------------------------|--------------------|
| Keynote speaker at "COSMO21", Chania, Greece                            | May 20, 2024       |
| UChicago DSI research highlight, Chicago, IL                            | May 16, 2024       |
| KICP Colloquium, KICP, Chicago, IL                                      | April 10, 2024     |
| Invited talk at UPenn cosmology group, UPenn, Pennsylvania, PA          | November 20, 2023  |
| Invited talk at "KIPAC@20", SLAC, Menlo Park, CA                        | September 12, 2023 |
| Invited talk at "Modified Gravity", UIUC, Urbana Champaign, IL          | May 19, 2023       |
| Overview talk, "LSS x CMB Workshop", Kyoto, Japan                       | April 10, 2023     |
| Invited talk at Kahn Symposium, SLAC, Menlo Park, CA                    | March 3, 2023      |
| Lecturer at "Cosmology on the Beach", Playa del Carmen, Mexico          | November 30, 2022  |
| Astronomy Seminar, University of Cincinnati, OH                         | November 1, 2022   |
| Physics Colloquium, University of Kentucky, Lexington, KY               | October 28, 2022   |
| Plenary talk at "DECam at 10 Years Workshop", Tucson, AZ                | September 12, 2022 |
| Discussion session at Higher-Order Statistics Workshop, Aspen, CO       | June 5, 2022       |
| Talk at CMB-S4 Collaboration Meeting, remote                            | March 13, 2022     |
| EPFL Colloquium, remote                                                 | March 7, 2022      |
| Astronomy and Astrophysics Colloquium, Chicago, IL                      | February 16, 2022  |
| CIERA Astronomy Seminar at Northwestern, Evanston, IL                   | November 9, 2021   |
| Plenary talk at "Rubin Project and Community Workshop", remote          | August 12, 2021    |
| Invited review talk "Growth of Structure Summer Seminar Series", remote | June 23, 2021      |
| CITA Seminar, remote                                                    | March 28, 2021     |
| Invited Panel for SLAC public talk series, remote                       | September 29, 2020 |
| Invited talk at IPMU conference "Cosmic Acceleration", Tokyo, Japan     | February 17, 2020  |
| Physics Colloquium, Kansas State University, Manhattan, KS              | October 21, 2019   |
| Invited talk at conference "Cosmic Controversy", Chicago, IL            | October 5, 2019    |
| Plenary talk at conference "COSMO19", Aachen, Germany                   | September 2, 2019  |
| Invited talk at "LSST DESC Dark Matter Workshop", Chicago, IL           | August 5, 2019     |
| Plenary talk at conference APS DPF, Boston, MA                          | July 31, 2019      |
| Astrophysics Seminar, McGill University, Montreal, Canada               | July 3, 2019       |
| Plenary talk at "Quantum Theory and Symmetry-XI", Montreal, Canada      | July 2, 2019       |
| Panelist at workshop ICG25, State College, PA                           | June 24, 2019      |
| Fermilab Astrophysics Seminar, Baltavia, IL                             | June 10, 2019      |
| Invited talk at LSST in South America, Sao Paulo, Brazil                | December 18, 2018  |
| Kavli Symposium, Oslo, Norway                                           | September 1, 2018  |
| Invited talk at COSPAR, Pasadena, CA, USA                               | July 16, 2018      |
| Invited talk at APS, Columbus, OH, USA                                  | April 17, 2018     |
| UChicago Astro Seminar, Chicago, IL, USA                                | February 27, 2018  |
| Rutgers Astro Seminar, New Brunswick, NJ, USA                           | February 19, 2018  |
| Duke Physics Colloquium, Durham, NC, USA                                | February 14, 2018  |
| University of Pittsburgh Physics Colloquium, Pittsburgh, PA, USA        | January 30, 2018   |
| UC Berkeley Physics Colloquium, Berkeley, CA, USA                       | January 23, 2018   |
| Cosmology Group Meeting, CCA, NY, USA                                   | November 9, 2017   |
| Cosmology Seminar, Princeton/IAS, NJ, USA                               | November 6, 2017   |
| Fermilab Astro Seminar, Fermilab, IL, USA                               | October 23, 2017   |
| Astro/Cosmology Seminar, CMU, PA, USA                                   | October 13, 2017   |

|                                                                 |                    |
|-----------------------------------------------------------------|--------------------|
| Astrophysics and Cosmology Seminar, UIUC, IL, USA               | September 20, 2017 |
| Cosmology Seminar, BNL, NY, USA                                 | September 14, 2017 |
| Instrumentation Seminar, BNL, NY, USA                           | September 13, 2017 |
| The Nonlinear Universe, Smartno, Slovenia                       | July 20, 2017      |
| Fermilab 50th User Meeting, Fermilab, IL, USA                   | June 8, 2017       |
| KICP Colloquium, KICP, IL, USA                                  | May 31, 2017       |
| Astronomy Chalk Talk, U of Chicago, IL, USA                     | January 24, 2017   |
| Cosmology Seminar, UCL, London, UK                              | December 21, 2016  |
| Astronomy Colloquium, UIUC, IL, USA                             | November 1, 2016   |
| KICP Friday Seminar, KICP, IL, USA                              | October 7, 2016    |
| Cosmology Seminar, KIPAC, CA, USA                               | May 16, 2016       |
| Kosmologietag Overview Talk, Bielefeld University, Germany      | April 29, 2016     |
| Astrophysics Colloquium, ASIAA, Taipei, Taiwan                  | March 28, 2016     |
| RAS Specialist Discussion Meeting, London, UK                   | February 12, 2016  |
| Swiss Python Summit, Rapperswil, Switzerland                    | February 5, 2016   |
| Astrophysics Seminar, Rutgers University, NJ, USA               | August 11, 2015    |
| Cosmology Lunch, Princeton University, NY, USA                  | August 10, 2015    |
| Fourteenth Marcel Grossmann Meeting (MG14) Rome, Italy          | July 17, 2015      |
| APS April meeting, Baltimore, MD, USA                           | April 14, 2015     |
| Weekly Colloquium, IEEC-CSIC, Barcelona, Spain                  | October 8, 2014    |
| Astrophysics Seminar, ASIAA, Taipei, Taiwan                     | September 19, 2014 |
| Research Seminar Shanghai Jiao Tong University, Shanghai, China | September 9, 2014  |
| DES-LSST Joint Workshop, Fermilab, IL, USA                      | March 24, 2014     |
| Swiss Cosmology Day, ETH Zurich, Switzerland                    | February 6, 2014   |
| ETH Research Seminar, ETH Zurich, Zurich, Switzerland           | September 19, 2013 |
| Astrophysics Seminar, JPL, CA, USA                              | September 2012     |
| Special Seminar, IPMU, Tokyo, Japan                             | August 2012        |
| SnowPAC, Snowbird, CO, USA                                      | March 22, 2012     |

## Publications – Chihway Chang

---

Lead author (marked with \* and bold numbers, including as first 1-3 authors or papers from students or postdocs I directly advise and have significant contributions) of 53+ refereed publications in weak gravitational lensing, cross-correlation, and other large-scale cosmology topics. Contributing author of a total of 187+ publications. Full publication list available at [ORCHID](#) and [ADS](#).

SUBMITTED  
JOURNAL  
PUBLICATIONS

187. C. To, T. Shin, **C. Chang**... et al., *Robust Photometry for Roman High-Latitude Imaging Survey Cosmology Using Roman and Rubin Imaging*. Arxiv e-print (2026) [2607.09849](#).
186. B. Remy, **C. Chang**\*, R. Willett, *Joint inference of weak lensing convergence map and cosmology with diffusion models*. Arxiv e-print (2026) [2606.31988](#).
185. A. Ouellette, Y. Omori, ...**C. Chang**... et al., *Cross-correlation of SPT-3G D1 CMB lensing and DES Y3 galaxy lensing*. Arxiv e-print (2026) [2606.26223](#).
184. J. Zhang, **C. Chang**\*... et al., *Accurate modeling for  $3\times 2$ pt analyses in Roman and Rubin: a study of model approximations*. Arxiv e-print (2026) [2606.23781](#).
183. Y. Omori, J. Zeghal, **C. Chang**\*... et al., *Towards Practical Field-Level Inference for Weak Lensing*. Arxiv e-print (2026) [2606.12255](#).
182. The Dark Energy Survey Collaboration, ...**C. Chang**... et al., *Constraints on Dynamical Dark Energy from Multiple Probes in the Full Dark Energy Survey*. Arxiv e-print (2026) [2605.27221](#).
181. S. Srinivasan, ...**C. Chang**... et al., *Brightest Cluster Galaxy ellipticity as proxy for halo shape: Orientation bias, assembly bias, and potential selection effects in SZ-selected clusters*. Arxiv e-print (2026) [2603.23689](#).
180. The Dark Energy Survey Collaboration, ...**C. Chang**... et al., *Dark Energy Survey Year 6 Results: Cosmological Constraints from Cosmic Shear*. Arxiv e-print (2026) [2602.10065](#).
179. D. Sanchez-Cid, ...**C. Chang**... et al., *Dark Energy Survey Year 6 Results: Weak Lensing and Galaxy Clustering Cosmological Analysis Framework*. Arxiv e-print (2026) [2601.14859](#).
178. The Dark Energy Survey Collaboration, ...**C. Chang**... et al., *Dark Energy Survey Year 6 Results: Cosmological Constraints from Galaxy Clustering and Weak Lensing*. Arxiv e-print (2026) [2601.14559](#).
177. LSST Dark Energy Science Collaboration, ...**C. Chang**... et al., *Opportunities in AI/ML for the Rubin LSST Dark Energy Science Collaboration*. Arxiv e-print (2026) [2601.14235](#).
176. T. Pedersen, ...**C. Chang**... et al., *Extracting intrinsic alignments in the Dark Energy Survey's year 1 data, using the self-calibration method and LSST-DESC tools*. Arxiv e-print (2026) [2601.10314](#).
175. T. Shin, ...**C. Chang**... et al., *Weak Lensing Mass Calibration of the ACT DR5 Galaxy Clusters with the DES Year 3 Weak Lensing Data*. Arxiv e-print (2025) [2512.18935](#).
174. B. Yin, ...**C. Chang**... et al., *Dark Energy Survey Year 6 Results: Redshift Calibration of the Weak Lensing Source Galaxies*. Arxiv e-print (2025) [2510.23566](#).
173. M. Rodriguez-Monroy, ...**C. Chang**... et al., *Dark Energy Survey Year 6 Results: improved mitigation of spatially varying observational systematics with masking*. Arxiv e-print (2025) [2509.07943](#).

172. S. Pandey, ...**C. Chang**... et al., *Constraints on cosmology and baryonic feedback with joint analysis of Dark Energy Survey Year 3 lensing data and ACT DR6 thermal Sunyaev-Zel'dovich effect observations*. Arxiv e-print (2025) 2506.07432.
171. A. Campos, ...**C. Chang**... et al., *Enhancing weak lensing redshift distribution characterization by optimizing the Dark Energy Survey Self-Organizing Map Photo-z method*. Arxiv e-print (2024) 2408.00922.
170. E. Krause, ...**C. Chang**... et al., *Dark Energy Survey Year 3 Results: Multi-Probe Modeling Strategy and Validation*. Arxiv e-print (2021) 2105.13548.
169. C. Davis, ...**C. Chang**... et al., *Dark Energy Survey Year 1 Results: Cross-Correlation Redshifts in the DES – Calibration of the Weak Lensing Source Redshift Distributions*. ArXiv e-prints (2017) 1710.02517.
168. E. Krause, ...**C. Chang**... et al., *Dark Energy Survey Year 1 Results: Multi-Probe Methodology and Simulated Likelihood Analyses*. ArXiv e-prints (2017) 1706.09359.
167. G. Giannini, ...**C. Chang**... et al., *Dark Energy Survey Year 6 results: Redshift calibration of the MagLim++ lens sample*. PRD **114**, 023520 (2026) 2509.07964.
166. C. To, **C. Chang**\*... et al., *First detection of weak lensing around spectroscopically confirmed low-mass galaxies*. PRD **114**, 023040 (2026) 2509.20458.
165. M. Gatti, D. Anbajagane, **C. Chang**\*... et al., *DECADE+DES Y3 weak lensing mass map: A 13,000 deg<sup>2</sup> view of cosmic structure from 270 million galaxies*. PRD **113**, L081303 (2026) 2509.03798.
164. G. Zacharegkas, **C. Chang**\*, J. Prat... et al., *Constraining the Stellar-to-Halo Mass Relation with Galaxy Clustering and Weak Lensing from DES Year 3 Data*. OJAp **9**, 55046 (2026).
163. G. Giannini, ...**C. Chang**... et al., *Dark Energy Survey Year 6 results: Galaxy-galaxy lensing*. PRD **113**, 103503 (2026) 2601.15175.
162. J. McCullough, ...**C. Chang**... et al., *Dark Energy Survey Year 3: Blue shear*. PRD **113**, 103509 (2026) 2410.22272.
161. W. d'Assignies, ...**C. Chang**... et al., *Dark Energy Survey Year 6 results: Clustering redshifts and importance sampling of self-organized-maps  $n(z)$  realizations for  $3 \times 2pt$  samples*. PRD **113**, 123549 (2026) 2510.23565.
160. E. Legnani, ...**C. Chang**... et al., *Dark Energy Survey year 6 results: Magnification modeling and its impact on galaxy clustering and galaxy-galaxy lensing cosmology*. PRD **113**, 123520 (2026) 2601.14833.
159. Vera C. Rubin Observatory Team, ...**C. Chang**... et al., *The Vera C. Rubin Observatory Data Preview I*. AJ **171**, 360 (2026) 2603.23786.
158. A. Thomsen, ...**C. Chang**... et al., *Dark Energy Survey Year 3 results: Simulation-based  $\Lambda$ CDM inference from weak lensing and galaxy clustering maps with deep learning: Analysis design*. PRD **113**, 083501 (2026) 2511.04681.
157. The Dark Energy Survey Collaboration, ...**C. Chang**... et al., *Dark Energy Survey: Implications for cosmological expansion models from the final DES baryon acoustic oscillation and supernova data*. PRD **113**, 063530 (2026) 2503.06712.
156. Q. Hang, ...**C. Chang**... et al., *Biasing from galaxy trough and peak profiles with the DES Y3 redMaGiC galaxies and the weak lensing mass map*. MNRAS **546**, stag006 (2026) 2509.18967.

155. C. Y. Tan, ...**C. Chang**... et al., *DELVE Milky Way Satellite Galaxy Census. I. Satellite Population and Survey Selection Function in DES, DELVE, and Pan-STARRS*. ApJ **1000**, 87 (2026) 2509.12313.
154. J. Prat, M. Gatti, C. Doux, P. Pranav, **C. Chang**... et al., *Dark Energy Survey Year 3 results:  $\Lambda$ CDM cosmology from simulation-based inference with persistent homology on the sphere*. MNRAS **545**, staf2152 (2026) 2506.13439.
153. K. Bechtol, ...**C. Chang**... et al., *Dark Energy Survey Year 6 Results: Photometric Data Set for Cosmology*. ApJS **282**, 62 (2026) 2501.05739.
152. M. Yamamoto, ...**C. Chang**... et al., *Dark Energy Survey Year 6 Results: Cell-based Coadds and Metadetection Weak Lensing Shape Catalogue*. MNRAS **543**, 4156 (2025) 2501.05665.
151. D. Anbajagane, **C. Chang**\*... et al., *The DECADE cosmic shear project IV: cosmological constraints from 107 million galaxies across  $5,400 \text{ deg}^2$  of the sky*. OJA **8** 46161 (2025) 2502.17677.
150. D. Anbajagane, **C. Chang**\*... et al., *The DECADE cosmic shear project III: validation of analysis pipeline using spatially inhomogeneous data*. OJA **8** 46160 (2025) 2502.17676.
149. D. Anbajagane, A. Alarcon, R. Teixeira, **C. Chang**\*... et al., *The DECADE cosmic shear project II: photometric redshift calibration of the source galaxy sample*. OJA **8** 46159 (2025) 2502.17675.
148. D. Anbajagane, **C. Chang**\*... et al., *The DECADE cosmic shear project I: A new weak lensing shape catalog of 107 million galaxies*. OJA **8** 46158 (2025) 2502.17674.
147. The Dark Energy Survey Collaboration, ...**C. Chang**\*... et al., *Dark Energy Survey Year 3 Results: Cosmological Constraints from Cluster Abundances, Weak Lensing, and Galaxy Clustering*. PRD **112**, 083535 (2025) 2503.13632.
146. J. Jefferson, Y. Omori, **C. Chang**\*... et al., *Reanalysis of Stage-III cosmic shear surveys: A comprehensive study of shear diagnostic tests*. OJA **8** 139 (2025) 2505.03964.
145. C. To, E. Krause, **C. Chang**\*... et al., *Dark Energy Survey: Modeling strategy for multiprobe cluster cosmology and validation for the Full Six-year Dataset*. PRD **112**, 063537 (2025) 2503.13631.
144. A. Gomes, ...**C. Chang**... et al., *Dark Energy Survey Year 3 Results: Cosmological constraints from second- and third-order shear statistics*. PRD **112**, 123515 (2025) 2508.14018.
143. B. Thakore, ...**C. Chang**... et al., *High-significance detection of correlation between the unresolved gamma-ray background and the large-scale cosmic structure*. JCAP **2025**, 6 (2025) 2501.10506.
142. D. Anbajagane, ...**C. Chang**... et al., *Dark Energy Survey Year 6 Results: Synthetic-source Injection Across the Full Survey Using Balrog*. OJA **8** 65 (2025) 2501.05683.
141. B. Levine, J. Sanchez, **C. Chang**\*... et al., *Galaxy Clustering with LSST: Effects of Number Count Bias from Blending*. OJA **8** 38 (2025) 2411.14564 .
140. Xianzhe. Tang, ...**C. Chang**... et al., *Uniting the Observed Dynamical Dark Energy Preference with the Discrepancies in  $\Omega_m$  and  $H_0$  across Cosmological Probes*. AJ Letter **983**, 1 (2025) 2412.04430 .
139. S. Bocquet, ...**C. Chang**... et al., *Multiprobe cosmology from the abundance of SPT clusters and DES galaxy clustering and weak lensing*. PRD **111** 063533 (2025) 2412.07765.

138. M. Gatti, ...**C. Chang**... et al., *Dark Energy Survey Year 3 results: simulation-based cosmological inference with wavelet harmonics, scattering transforms, and moments of weak lensing mass maps II. Cosmological results*. PRD **111** 6 (2025) 2405.10881.
137. T. Schutt, ...**C. Chang**... et al., *Dark Energy Survey Year 6 Results: Point-Spread Function Modeling*. OJA **8** 26 (2025) 2501.05781.
136. C. Y. Tan, ...**C. Chang**... et al., *A Pride of Satellites in the Constellation Leo? Discovery of the Leo VI Milky Way Satellite Galaxy with DELVE Early Data Release 3*. AJ **979** 2 (2025) 2408.00865.
135. L. Faga, ...**C. Chang**... et al., *Dark Energy Survey Year 3 Results: Cosmology from galaxy clustering and galaxy-galaxy lensing in harmonic space*. MNRAS **536** 2 (2025) 2406.12675.
134. N. Jeffrey, ...**C. Chang**... et al., *Dark Energy Survey Year 3 results: likelihood-free, simulation-based  $\Lambda$ CDM inference with neural compression of weak-lensing map statistics*. MNRAS **526** 2 (2025) 2403.02314.
133. L. E. Gagnon, D. Anbajagane, J. Prat, **C. Chang\***, J. Frieman, *Cosmological Constraints from Combining Galaxy Surveys and Gravitational Wave Observatories*. OJA **7** 110 (2024) 2312.16289.
132. D. Anbajagane, S. Pandey, **C. Chang\*** *Map-level baryonification: Efficient modelling of higher-order correlations in the weak lensing and thermal Sunyaev-Zeldovich fields*. OJA **7** 110 (2024) 2409.03822.
131. S. Bocquet, ...**C. Chang**... et al., *SPT Clusters with DES and HST Weak Lensing. II. Cosmological Constraints from the Abundance of Massive Halos*. PRD **110** 8 (2024) 2401.02075.
130. S. Bocquet, ...**C. Chang**... et al., *SPT clusters with DES and HST weak lensing. I. Cluster lensing and Bayesian population modeling of multiwavelength cluster datasets*. PRD **110** 8 (2024) 2310.12213.
129. N. Chiocone, J. Prat, G. Zacharegkas, **C. Chang\***... et al., *Weak Gravitational Lensing around Low Surface Brightness Galaxies in the DES Year 3 Data*. OJA **7** 89 (2024) 2407.19081.
128. L. Bigwood, ...**C. Chang**... et al., *Weak lensing combined with the kinetic Sunyaev Zel'dovich effect: A study of baryonic feedback*. MNRAS **534** 1 (2024) 2404.06098.
127. DES, ...**C. Chang**... et al., *Dark Energy Survey: A 2.1% measurement of the angular Baryonic Acoustic Oscillation scale at redshift  $z_{\text{eff}} = 0.85$  from the final dataset*. PRD **110** 6 (2024) 2402.10696.
126. DES, ...**C. Chang**... et al., *The Dark Energy Survey: Cosmology Results With 1500 New High-redshift Type Ia Supernovae Using The Full 5-year Dataset*. AJ Letter **973** 1 (2024) 2401.02929.
125. S. Grandis, ...**C. Chang**... et al., *The SRG/eROSITA All-Sky Survey: Dark Energy Survey Year 3 Weak Gravitational Lensing by eRASS1 selected Galaxy Clusters*. A&A **687**, 178 (2024) 2402.08455.
124. D. Anbajagane, **C. Chang\***, H. Lee, M. Gatti, *Primordial non-Gaussianities with weak lensing: information on non-linear scales in the ULAGAM full-sky simulations*. JCAP **2024**, 3 (2024) 2310.02349.
123. M. Lin, ...**C. Chang**... et al., *Late time modification of structure growth and the  $S_8$  tension*. PRD **109**, 6 (2024) 2308.16183.
122. M. Gatti, ...**C. Chang**... et al., *Dark Energy Survey Year 3 results: Simulation-based cosmological inference with wavelet harmonics, scattering transforms, and moments of weak lensing mass maps. Validation on simulations*. PRD **109**, 6 (2024) 2310.17557.

121. S. Shaikh, ...**C. Chang**... et al., *Cosmology from cross-correlation of ACT-DR4 CMB lensing and DES-Y3 cosmic shear*. MNRAS **528**, 2112 (2024) 2309.04412.
120. M. Gatti, ...**C. Chang**... et al., *Detection of the significant impact of source clustering on higher-order statistics with DES Year 3 weak gravitational lensing data*. MNRAS **527**, 115 (2024) 2307.13860.
119. D. Anbajagane, **C. Chang**\* ... et al., *Cosmological shocks around galaxy clusters: a coherent investigation with DES, SPT, and ACT*. MNRAS **527**, 9378 (2023) 2310.00059.
118. Dark Energy Survey; Kilo-Degree Survey Collaborations, T.M.C. Abbott...**C. Chang**... et al., *DES Y3 + KiDS-1000: Consistent cosmology combining cosmic shear surveys*. OJA **6**, 36 (2023) 2305.17173.
117. C. Zhou,...**C. Chang**... et al., *The Intrinsic Alignment of Red Galaxies in DES Y1 redMaPPer Galaxy Cluster*. MNRAS **526**, 323 (2023) 2302.12325.
116. Y. Zhang, ...**C. Chang**, *The Effect of Splashback on Weak Lensing Mass Estimates of Galaxy Clusters and Groups*. OJA **6** 46 (2023) 2212.05406.
115. C. Sanchez, ...**C. Chang**... et al., *The Dark Energy Survey Year 3 high redshift sample: Selection, characterization and analysis of galaxy clustering*. MNRAS **525** 3 (2023) 2211.16593.
114. Z. Zhang, Y. Omori, **C. Chang**\*, *Detecting deviations from Gaussianity in high-redshift CMB lensing maps*. MNRAS **524** 6392 (2022) 2211.09617.
113. A. Chen, ...**C. Chang**... et al., *Constraining the Baryonic Feedback with Cosmic Shear Using the DES Year-3 Small-Scale Measurements*. MNRAS **518** 5340 (2023) 2206.08591.
112. D. Anbajagane, **C. Chang**\* ... et al., *Beyond the 3rd moment: A practical study of using lensing convergence CDFs for cosmology with DES Y3*. MNRAS **526**, 5530 (2023) 2308.03863.
111. S. Samuroff, ...**C. Chang**... et al., *The Dark Energy Survey Year 3 and eBOSS: constraining galaxy intrinsic alignments across luminosity and colour space*. MNRAS **524**, 2195 (2023) 2212.11319
110. J. Elvin-Poole, ...**C. Chang**... et al., *Dark Energy Survey Year 3 results: magnification modelling and impact on cosmological constraints from galaxy clustering and galaxy-galaxy lensing*. MNRAS **523**, 3649 (2023) 2209.09782.
109. J. M. Shao, D. Anbajagane, **C. Chang**\*, *Baryonic imprints on DM haloes: the concentration-mass relation in the CAMELS simulations*. MNRAS **523**, 3258 (2023) 2212.05964.
108. J. Sanchez, Y. Omori, **C. Chang**\* ... et al., *Mapping gas around massive galaxies: cross-correlation of DES Y3 galaxies and Compton-y maps from SPT and Planck*. MNRAS **522**, 3163 (2023) 2210.08633.
107. J. Prat, ...**C. Chang**... et al., *Non-local contribution from small scales in galaxy-galaxy lensing: comparison of mitigation schemes*. MNRAS **522**, 412 (2023) 2212.03734.
106. J. Prat, J. Zuntz, Y. Omori, **C. Chang**\* ... et al., *The catalog-to-cosmology framework for weak lensing and galaxy clustering for LSST*. OJAp **6**, 13 (2023) 2212.09345.
105. E. P. Longley, **C. Chang**\*, C. W. Walter, J. Zuntz... et al., *A Unified Catalog-level Re-analysis of Stage-III Cosmic Shear Surveys*. MNRAS **520**, 5016 (2023) 2208.07179.
104. The Dark Energy Survey Collaboration, ...**C. Chang**... et al., *Dark Energy Survey Year 3 Results: Constraints on extensions to  $\Lambda$ CDM with weak lensing and galaxy clustering*. PRD **107**, 3504 (2023) 2207.05766.

103. The Dark Energy Survey and the South Pole Telescope Collaborations, ...**C. Chang**\* ... et al., *Joint analysis of DES Year 3 data and CMB lensing from SPT and Planck III: Combined cosmological constraints*. PRD **107**, 3531 (2023) [2206.10824](#).
102. **C. Chang**\*, Y. Omori, E. J. Baxter, ... et al., *Joint analysis of DES Year 3 data and CMB lensing from SPT and Planck II: Cross-correlation measurements and cosmological constraints*. PRD **107**, 3530 (2023) [2203.12440](#).
101. Y. Omori, E. J. Baxter, **C. Chang**\*, ... et al., *Joint analysis of DES Year 3 data and CMB lensing from SPT and Planck I: Construction of CMB Lensing Maps and Modeling Choices*. PRD **107**, 3529 (2023) [2203.12439](#).
100. M. Gatti, B. Jain, **C. Chang**... et al., *Dark Energy Survey Year 3 results: Cosmology with moments of weak lensing mass maps*. PRD **106**, 3509 (2021) [2110.10141](#).
99. A. Drlica-Wagner, ... **C. Chang**... et al., *The DECam Local Volume Exploration Survey Data Release 2*. AJS **261**, 38 (2022) [2203.16565](#).
98. C. Doux, ... **C. Chang**... et al., *Dark energy survey year 3 results: cosmological constraints from the analysis of cosmic shear in harmonic space*. MNRAS **515**, 1942 (2022) [2203.07128](#).
97. L. Secco, M. Jarvis, B. Jain, **C. Chang**... et al., *Dark Energy Survey Year 3 Results: Three-Point Shear Correlations and Mass Aperture Moments*. PRD **105**, 10 (2022) [2201.05227](#).
96. A. Kovacs, ...**C. Chang**... et al., *The DES view of the Eridanus supervoid and the CMB Cold Spot*. MNRAS **510**, 216 (2022) [2112.07699](#).
95. D. Anbajagane, **C. Chang**\*, B. Jain... et al., *Shocks in the Stacked Sunyaev-Zel'dovich Profiles of Clusters II: Measurements from SPT-SZ + Planck Compton-y Map*. MNRAS **514**, 1645 (2022) [2111.04778](#).
94. Z. Zhang, **C. Chang**\*, *Transitioning from Stage-III to Stage-IV: Cosmology from galaxy×CMB lensing and shear×CMB lensing*. MNRAS **514**, 2181 (2022) [2111.04917](#).
93. J. Prat, C. Hogan, **C. Chang**, J. Frieman, *Vacuum Energy Density Measured from Cosmological Data*. JCAP **06**, 015 (2022) [2111.08151](#).
92. D. Zuercher, ...**C. Chang**... et al., *Dark Energy Survey Year 3 results: Cosmology with peaks using an emulator approach*. MNRAS **511**, 2075 (2022) [2110.10135](#).
91. P. Fiedorowicz, ...**C. Chang**... et al., *KaRMMa – Kappa Reconstruction for Mass Mapping*. MNRAS **512**, 73 (2022) [2105.14699](#).
90. E. Kovac, ...**C. Chang**... et al., *Validating Synthetic Galaxy Catalogs for Dark Energy Science in the LSST Era*. OJAp **5** 1 (2021) [2110.03769](#).
89. M. Gatti, ...**C. Chang**... et al., *Cross-correlation of DES Y3 lensing and ACT+Planck thermal Sunyaev Zel'dovich Effect I: Measurements, systematics tests, and feedback model constraints*. PRD **105**, 123525 (2022) [2108.01600](#).
88. S. Pandey, ...**C. Chang**... et al., *Cross-correlation of DES Y3 lensing and ACT+Planck thermal Sunyaev Zel'dovich Effect II: Modeling and constraints on halo pressure profiles*. PRD **105**, 123526 (2022) [2108.01601](#).
87. The Dark Energy Survey Collaboration, ...**C. Chang**... et al., *Dark Energy Survey Year 3 Results: Cosmological Constraints from Galaxy Clustering and Weak Lensing*. PRD **105**, 023520 (2022) [2105.13549](#).
86. J. DeRose, ...**C. Chang**... et al., *Dark Energy Survey Year 3 results: Cosmology from combined galaxy clustering and lensing validation on cosmological simulations*. PRD **105**, 123520 (2021) [2105.13547](#).

85. S. Pandey, ...**C. Chang**... et al., *Dark Energy Survey Year 3 Results: Constraints on cosmological parameters and galaxy bias models from galaxy clustering and galaxy-galaxy lensing using the redMaGiC sample*. PRD **106**, 043520 (2022) [2105.13545](#).
84. L. Secco, ...**C. Chang**... et al., *Dark Energy Survey Year 3 Results: Cosmology from Cosmic Shear and Robustness to Modeling Uncertainty*. PRD **105**, 023515 (2022) [2105.13544](#).
83. A. Amon, ...**C. Chang**... et al., *Dark Energy Survey Year 3 Results: Cosmology from Cosmic Shear and Robustness to Data Calibration*. PRD **105**, 023514 (2022) [2105.13543](#).
82. C. Sanchez, ...**C. Chang**... et al., *Dark Energy Survey Year 3 Results: Exploiting small-scale information with lensing shear ratios*. PRD **105** 083529 (2022) [2105.13542](#).
81. J. Prat, ...**C. Chang**... et al., *Dark Energy Survey Year 3 Results: High-precision measurement and modeling of galaxy-galaxy lensing*. PRD **105**, 083528 (2022) [2105.13541](#).
80. O. Friedrich, ...**C. Chang**... et al., *Dark Energy Survey Year 3 Results: Covariance Modelling and its Impact on Parameter Estimation and Quality of Fit*. MNRAS **508**, 3125 (2021) [2012.08568](#).
79. G. Zacharegkas, **C. Chang**\*, J. Prat... et al., *Dark Energy Survey Year 3 results: Galaxy-halo connection from galaxy-galaxy lensing*. MNRAS **509**, 3119 (2022) [2106.08438](#).
78. T. Shin, B. Jain, S. Adhikari, E. J. Baxter, **C. Chang**... et al., *The mass and galaxy distribution around SZ-selected clusters*. MNRAS **507**, 5758 (2021) [2105.05914](#).
77. S. Lee, ...**C. Chang**... et al., *Galaxy-galaxy lensing with the DES-CMASS catalogue: measurement and constraints on the galaxy-matter cross-correlation*. MNRAS **509**, 2033 (2022) [2104.11319](#).
76. E. Baxter, S. Adhikari, J. Vega-Ferrero, W. Cui, **C. Chang**... et al., *Shocks in the Stacked Sunyaev-Zel'dovich Profiles of Clusters I: Analysis with the Three Hundred Simulations*. MNRAS **508**, 1777 (2021) [2101.04179](#).
75. S. Adhikari, ...**C. Chang**... et al., *Probing Galaxy Evolution in Massive Clusters using ACT and DES: Splashback as a Cosmic Clock*. ApJ **923**, 37 (2021) [2008.11663](#).
74. F. Andrade-Oliveira, ...**C. Chang**... et al., *Galaxy clustering in harmonic space from the dark energy survey year 1 data: compatibility with real-space results*. MNRAS **505**, 5714 (2021) [2103.14190](#).
73. N. Jeffrey, M. Gatti, **C. Chang**, L. Whiteway, U. Demirbozan... et al., *Dark Energy Survey Year 3 results: Curved-sky weak lensing mass map reconstruction*. MNRAS **505**, 4626 (2021) [2105.13539](#).
72. The Dark Energy Survey Collaboration, ...**C. Chang**... et al., *The Dark Energy Survey Data Release 2*. ApJS **255**, 20 (2021) [2101.05765](#).
71. M. Gatti, ...**C. Chang**... et al., *Dark energy survey year 3 results: weak lensing shape catalogue*. MNRAS **504**, 4312 (2021) [2011.03408](#).
70. **C. Chang**, A. Drlica-Wagner, S. M. Kent, D. M. Wang, M. H. L. S. Wang, *A machine learning approach to the detection of ghosting and scattered light artifacts in dark energy survey images*. Astronomy and Computing **36**, 100474 (2021) [2105.10524](#).
69. I. Sevilla-Noarbe, ...**C. Chang**... et al., *Dark Energy Survey Year 3 Results: Photometric Data Set for Cosmology*. ApJS **254**, 24 (2021) [2011.03407](#).
68. P. Lemos, M. Raveri, A. Campos, Y. Park, **C. Chang**\* ... et al., *Assessing tension metrics with Dark Energy Survey and Planck data*. MNRAS **505**, 6179 (2021) [2012.09554](#).

67. C. Doux, **C. Chang\***, B. Jain, J. Blazek, H. Camacho... et al., *Consistency of cosmic shear analyses in harmonic and real space*. MNRAS **503**, 3796 (2021) [2011.06469](#).
66. C. Doux, E. Baxter, P. Lemos, **C. Chang**, A. Alarcon... et al., *Dark Energy Survey internal consistency tests of the joint cosmological probes analysis with posterior predictive distributions*. MNRAS **503**, 2688 (2021) [2011.03410](#).
65. C. To, ...**C. Chang**... et al., *Dark Energy Survey Year 1 Results: Cosmological Constraints from Cluster Abundances, Weak Lensing, and Galaxy Correlations*. PRL **126**, 141301 (2020) [2010.01138](#).
64. H. Sampaio-Santos, ...**C. Chang**... et al., *Is diffuse intracluster light a good tracer of the galaxy cluster matter distribution?*. MNRAS (2020) [2005.12275](#)
63. M. Jarvis, ...**C. Chang**... et al., *Dark Energy Survey Year 3 Results: Point Spread Function Modelling*. MNRAS **501**, 1282 (2021) [22011.03409](#)
62. A. Palmese, ...**C. Chang**... et al., *A Statistical Standard Siren Measurement of the Hubble Constant from the LIGO/Virgo Gravitational Wave Compact Object Merger GW190814 and Dark Energy Survey Galaxies*. ApJ **900**, 33 (2020) [2006.14961](#).
61. D. Tanoglidis, ...**C. Chang**... et al., *Shadows in the Dark: Low-Surface-Brightness Galaxies Discovered in the Dark Energy Survey*. ApJS **252**, 18 (2021) [2006.04294](#)
60. W. Hartley, **C. Chang\***, S. Samani, A. Carnero Rosell, T. Davis et al., *The Impact of Spectroscopic Incompleteness in Direct Calibration of Redshift Distributions for Weak Lensing Surveys*. MNRAS **496**, 4769 (2020) [2003.10454](#).
59. M. Gatti, **C. Chang\***, O. Friedrich, B. Jain, D. Bacon et al., *Dark Energy Survey Year 3 Results: Cosmology with Moments of Weak Lensing Mass Maps – Validation on Simulations*. MNRAS **498**, 4060 (2020) [1911.05568](#).
58. B. Mawdsley, D. Bacon, **C. Chang**, P. Melchior, E. Rozo et al., *Dark Energy Survey Year 1 Results: Wide-field Mass Maps via Forward Fitting in Harmonic Space*. MNRAS **493**, 5662 (2020) [1905.12682](#).
57. The Dark Energy Survey Collaboration, ...**C. Chang**... et al., *Dark Energy Survey Year 1 Results: Cosmological Constraints from Cluster Abundances and Weak Lensing*. PRD **102**, 023509 (2020). [2002.11124](#)
56. D. Tanoglidis, **C. Chang\***, J. Frieman, *Optimizing Galaxy Samples for Clustering Measurements in Photometric Surveys*. MNRAS **491**, 3535 (2020) [1908.07150](#).
55. Y. Fang, ...**C. Chang**... et al., *Dark Energy Survey Year 1 results: The Relationship between Mass and Light around Cosmic Voids*. MNRAS **490**, 3573 (2019) [1909.01386](#).
54. D. Korytov, ...**C. Chang**... et al., *CosmoDC2: A Synthetic Sky Catalog for Dark Energy Science with LSST*. ApJS **254**, 26 (2019) [1907.06530](#).
53. K. Vahi, M. Wang, **C. Chang** et al., *Workflows using Pegasus: Enabling Dark Energy Survey Pipelines*. ASPC **523**, 689 (2019).
52. The Dark Energy Survey Collaboration, ...**C. Chang\***... et al., *Dark Energy Survey Year 1 Results: Joint Analysis of Galaxy Clustering, Galaxy Lensing, and CMB Lensing Two-point Functions*. PRD **100**, 3541 (2019) [1810.02322](#).
51. Y. Omori, T. Giannantonio, A. Porredon, E. Baxter, **C. Chang** et al., *Dark Energy Survey Year 1 Results: tomographic cross-correlations between DES galaxies and CMB lensing from SPT+Planck*. PRD **100**, 3501 (2019) [1810.02342](#).
50. Y. Omori, E. Baxter, **C. Chang\***, D. Kirk, A. Alarcon et al., *Dark Energy Survey Year 1 Results: Cross-correlation between DES Y1 galaxy weak lensing and SPT+Planck CMB weak lensing*. PRD **100**, 3517 (2019) [1810.02441](#).

49. The Dark Energy Survey Collaboration, ...**C. Chang**... et al., *Cosmological Constraints from Multiple Probes in the Dark Energy Survey*. PRD **122**, 171301 (2019) 1811.02375.
48. The Dark Energy Survey Collaboration, ...**C. Chang**... et al., *Dark Energy Survey Year 1 Results: Constraints on Extended Cosmological Models from Galaxy Clustering and Weak Lensing*. PRD **99**, 123505 (2019) 1810.02499.
47. J. Prat, E. Baxter, T. Shin, C. Sanchez, **C. Chang** et al., *Cosmological lensing ratios with DES Y1, SPT and Planck*. MNRAS **487**, 1363 (2019) 1810.02212.
46. T. Shin, S. Adhikari, E. Baxter, **C. Chang**, B. Jain et al., *Measurement of the Splashback Feature around SZ-selected Galaxy Clusters with DES, SPT and ACT*. MNRAS **487**, 2900 (2019) 1811.06081.
45. **C. Chang**<sup>\*</sup>, M. Wang, S. Dodelson, T. Eifler, C. Heymans et al., *A Unified Analysis of Four Cosmic Shear Surveys*. MNRAS **482**, 3696 (2019) 1808.07335.
44. M. Fagioli, ...**C. Chang**... et al., *Forward Modeling of Spectroscopic Galaxy Surveys: Application to SDSS*. JCAP **11**, 015 (2018) 1803.06343.
43. E. Baxter, Y. Omori, **C. Chang**<sup>\*</sup>, T. Giannantonio, D. Kirk et al., *Dark Energy Survey Year 1 Results: Methodology and Projections for Joint Analysis of Galaxy Clustering, Galaxy Lensing, and CMB Lensing Two-point Functions*. PRD **99**, 023508 (2019) 1802.05257.
42. The Dark Energy Survey Collaboration, ...**C. Chang**... et al., *The Dark Energy Survey Data Release 1*. ApJS **239**, 18 (2018) 1801.03181.
41. R. Cawthon, ...**C. Chang**... et al., *Dark Energy Survey Year 1 Results: Calibration of redMaGiC Redshift Distributions in DES and SDSS from Cross-Correlations*. MNRAS **481**, 2427 (2018) 1712.07298.
40. **C. Chang**<sup>\*</sup>, E. Baxter, B. Jain, C. Sanchez, S. Adhikari et al., *The Splashback Feature around DES Galaxy Clusters: Galaxy Density and Weak Lensing Profiles*. ApJ **864**, 83 (2018) 1710.06808.
39. The Dark Energy Survey Collaboration, ...**C. Chang**... et al., *Dark Energy Survey Year 1 Results: Cosmological Constraints from Galaxy Clustering and Weak Lensing*. PRD **98**, 043526 (2018) 1708.01530.
38. J. Prat, ...**C. Chang** et al., *Dark Energy Survey Year 1 Results: Galaxy-Galaxy Lensing*. PRD **98**, 042005 (2018) 1708.01537.
37. M. Troxel, ...**C. Chang**... et al., *Dark Energy Survey Year 1 Results: Cosmological Constraints from Cosmic Shear*. PRD **98**, 043528 (2018) 1708.01538.
36. M. Troxel, E. Krause, **C. Chang**, T. F. Eifler, O. Friedrich et al., *Survey Geometry and the Internal Consistency of Recent Cosmic Shear Measurements*. MNRAS **476**, 4998 (2018) 1804.10663.
35. N. Jeffrey, ... **C. Chang**..., et al., *Improving Weak Lensing Mass Map Reconstructions using Gaussian and Sparsity Priors: Application to DES SV*. MNRAS **479**, 2871 (2018) 1801.08945.
34. O. Friedrich, ...**C. Chang**... et al., *Density Split Statistics: Joint Model of Counts and Lensing in Cells*. PRD **98**, 023508 (2018) 1710.05162.
33. D. Gruen, ...**C. Chang**... et al., *Density Split Statistics: Cosmological Constraints from Counts and Lensing in Cells in DES Y1 and SDSS*. PRD **98**, 023507 (2018) 1710.05045.
32. B. Hoyle, ...**C. Chang**... et al., *Dark Energy Survey Year 1 Results: Redshift Distributions of the Weak Lensing Source Galaxies*. MNRAS **478**, 592 (2018) 1708.01532.

31. M. Gatti, ...**C. Chang**... et al., *Dark Energy Survey Year 1 Results: Cross-Correlation Redshifts - Methods and Systematics Characterization*. MNRAS (2018) [1709.00992](#).
30. **C. Chang**\*, A. Pujol, B. Mawdsley, D. Bacon, J. Elvin-Poole, et al., *Dark Energy Survey Year 1 Results: Curved-Sky Weak Lensing Mass Map*. MNRAS **475**, 3165 (2018) [1708.01535](#).
29. E. Baxter, **C. Chang**\*, B. Jain, S. Adhikari, N. Dalal et al., *The Halo Boundary of Galaxy Clusters in the SDSS*. ApJ **841**, 18 (2017) [1702.01722](#).
28. J. Akeret, **C. Chang**\*, A. Lucchi, A. Refregier, *Radio Frequency Interference Mitigation using Deep Convolutional Neural Networks*. A&C **18**, 35–39 (2017) [1609.09077](#).
27. J. Akeret, S. Seehars, **C. Chang**, C. Monstein, A. Amara, A. Refregier, *HIDE & SEEK: End-to-End Packages to Simulate and Process Radio Survey Data*. A&C **18**, 8–17 (2017) [1607.07443](#).
26. **C. Chang**\*, C. Monstein, J. Akeret, S. Seehars, A. Refregier et al., *An Integrated System at the Bleien Observatory for Mapping the Galaxy*. MNRAS **464**, 1727–1737 (2017) [1607.07451](#).
25. N. MacCrann, ...**C. Chang**... et al., *Inference from the Small Scales of Cosmic Shear with Current and Future Dark Energy Survey Data*. MNRAS **465**, 2567–2583 (2017). [1608.01838](#).
24. L. Clerkin, ...**C. Chang**... et al., *Testing the Lognormality of the Galaxy and Weak Lensing Convergence Distributions from Dark Energy Survey Maps*. MNRAS **466**, 1444–1461 (2017). [1605.02036](#).
23. T. Kacprzak, ... **C. Chang**..., et al., *Cosmology Constraints from Shear Peak Statistics in Dark Energy Survey Science Verification Data*. MNRAS **463**, 3653–3673 (2016), [1603.05040](#).
22. B. Nord, ... **C. Chang**..., et al., *SPOKES: An End-to-End Simulation Facility for Spectroscopic Cosmological Surveys*. A&C **15**, 1–15 (2016), [1602.01480](#).
21. D. Kirk, Y. Omori, A. Benoit-Levy, R. Cawton, **C. Chang** et al., *Cross-correlation of Gravitational Lensing from DES Science Verification, SPT and Planck*. MNRAS **459**, 21 (2016), [1512.04535](#).
20. A. Pujol, **C. Chang**\*, E. Gazganaga, A. Amara, A. Refregier et al., *A New Method to Measure Galaxy Bias from the Density and Weak Lensing Fields*. MNRAS **462**, 35–47 (2016) [1601.00160](#).
19. **C. Chang**\*, A. Pujol, E. Gazganaga, A. Amara, A. Refregier et al., *Galaxy Bias from the DES Science Verification Data: Combining Galaxy Density Maps and Weak Lensing Maps*. MNRAS **459**, 3203 (2016), [1601.00405](#).
18. The Dark Energy Survey Collaboration .. **C. Chang**..., et al., *Cosmology from Cosmic Shear with DES Science Verification Data*. PRD **94**, 022001 (2016), [1507.05552](#).
17. M. Jarvis, ... **C. Chang**..., et al., *The DES Science Verification Weak Lensing Shear Catalogs*. MNRAS **460**, 2245 (2016), [1507.05603](#).
16. M.R. Becker, ... **C. Chang**..., et al., *Cosmic Shear Measurements with DES Science Verification Data*. PRD **94**, 022002 (2016), [1507.05598](#).
15. B. Leistedt, ... **C. Chang**..., et al., *Mapping and Simulating Systematics due to Spatially-Varying Observing Conditions in DES Science Verification Data*. ApJS **226**, 24 (2016), [1507.05647](#).
14. C. Bonnett, ... **C. Chang**..., et al., *Redshift Distributions of Galaxies in the DES Science Verification Shear Catalogue and Implications for Weak Lensing* PRD **94**, 042005 (2016), [1507.05909](#).

13. C. Bruderer, **C. Chang\***, A. Refregier, A. Amara, J. Berge et al., *Calibrated Ultra Fast Image Simulations for the Dark Energy Survey*. *ApJ* **817**, 25 (2016), [1504.02778](#).
12. **C. Chang\***, C. Monstein, A. Refregier, A. Amara, A. Glauser et al., *Beam Calibration of Radio Telescopes with Drones*. *PASP* **127**, 1131–1143, (2015), [1505.05885](#).
11. **C. Chang\***, V. Vikram, B. Jain, D. Bacon, A. Amara et al., *Wide-Field Lensing Mass Maps from DES Science Verification Data*. *PRL* **115**, 051301 (2015), [1505.01871](#).
10. V. Vikram, **C. Chang\***, B. Jain, D. Bacon, A. Amara et al., *Wide-Field Lensing Mass Maps from DES Science Verification Data: Methodology and Detailed Analysis*. *PRD* **92**, 022006 (2015), [1504.03002](#).
9. J.R. Peterson, ... **C. Chang**... et al., *Simulation of Astronomical Images from Optical Survey Telescopes using a Comprehensive Photon Monte Carlo Approach*. *ApJS* **218**, 14 (2015), [1504.06570](#).
8. **C. Chang\***, M.T. Busha, R.H. Wechsler, A. Refregier, A. Amara et al., *Modelling the Transfer Function for the Dark Energy Survey*. *ApJ* **801**, 73 (2015), [1411.0032](#).
7. **C. Chang\*** and B. Jain, *Delensing Galaxy Surveys*. *MNRAS* **443**, 102 (2014), [1405.1432](#).
6. R. Mandelbaum, B. Rowe, J. Bosch, **C. Chang**, F. Courbin et al., *The Third Gravitational Lensing Accuracy Testing (GREAT3) Challenge Handbook*. *ApJS* **212**, 5 (2014), [1308.4982](#).
5. **C. Chang\***, M. Jarvis, B. Jain, S.M. Kahn, D. Kirkby et al., *The Effective Number Density of Galaxies for Weak Lensing Measurements in the LSST Project*. *MNRAS* **434**, 2121 (2013), [1305.0793](#).
4. D. Bard, J.M. Kratochvil, **C. Chang**, M. May, S.M. Kahn et al., *Effect of Measurement Errors on Predicted Cosmological Constraints from Shear Peak Statistics with LSST*. *ApJ* **774**, 49 (2013), [1301.0830](#).
3. **C. Chang\***, S.M. Kahn, J.G. Jernigan, J.R. Peterson, Y. AlSayyad et al., *Spurious Shear in Weak Lensing with LSST*. *MNRAS* **428**, 2695 (2013), [1206.1378](#).
2. **C. Chang\***, P.J. Marshall, J.G. Jernigan, J.R. Peterson, S.M. Kahn et al., *Atmospheric PSF Interpolation for Weak Lensing in Short Exposure Imaging Data*. *MNRAS* **427**, 2572 (2012), [1206.1383](#).
1. J. Singal, R. Schindler, **C. Chang**, P. Czodrowski, and P. Kim, *A Multi-Chamber System for Analyzing the Outgassing, Deposition, and Associated Optical Degradation Properties of Materials in a Vacuum*. *Review of Scientific Instruments* **81**, 025101 (2010), [0910.4198](#).

ARXIV E-PRINTS,  
CONFERENCE  
PROCEEDINGS,  
POSTERS

12. E. J. Baxter, **C. Chang**, A. Hearin..., et al., *Snowmass2021: Opportunities from Cross-survey Analyses of Static Probes*. *ArXiv e-prints* (2022) [2203.06795](#).
11. D. Scolnic, ... **C. Chang**..., et al., *Optimizing the LSST Observing Strategy for Dark Energy Science: DESC Recommendations for the Deep Drilling Fields and other Special Programs*. *ArXiv e-prints* (2018) [1812.00516](#).
10. M. Lochner, ... **C. Chang**..., et al., *Optimizing the LSST Observing Strategy for Dark Energy Science: DESC Recommendations for the Wide-Fast-Deep Survey*. *ArXiv e-prints* (2018) [1812.00515](#).
9. The LSST Dark Energy Science Collaboration, *LSST Dark Energy Science Collaboration*. *ArXiv e-prints* (2012) [1211.0310](#).
8. J.R. Peterson, ... **C. Chang**..., et al., *LSST Image Simulations*. *American Astronomical Society Meeting Abstracts*, **219**, (2012).

7. A. Bradshaw, ... **C. Chang**..., et al., *LSST Probes of Dark Energy: New Energy vs New Gravity*. American Astronomical Society Meeting Abstracts, **219**, (2012).
6. R.R. Gibson, ... **C. Chang**..., et al., *A Framework for End to End Simulations of the Large Synoptic Survey Telescope*. Astronomical Data Analysis Software and Systems XX, **442**, p.329, (2011).
5. J. Pizagno, ... **C. Chang**..., et al., *Strong Lenses with LSST: Simulated 10-year Movies of Multiply-Imaged Quasars*. American Astronomical Society Meeting Abstracts, **217**, (2011).
4. **C. Chang**, S.M Kahn, G. Jernigan, J.R. Peterson, A. Rasmussen et al., *Shear Systematics in LSST Simulated Images*. American Astronomical Society Meeting Abstracts, **217**, (2011).
3. G. Jernigan, ... **C. Chang**..., et al., *Strong Lenses with LSST: Simulated 10-year Movies of Multiply-Imaged Quasars*. American Astronomical Society Meeting Abstracts, **217**, (2011).
2. K.S. Krughoff, ... **C. Chang**..., et al., *Strong Lenses with LSST: Simulated 10-year Movies of Multiply-Imaged Quasars*. American Astronomical Society Meeting Abstracts, **217**, (2011).
1. A. Connolly, ... **C. Chang**..., et al., *Simulating the LSST system*. Society of Photo-Optical Instrumentation Engineers (SPIE) Conference Series **7738**, p.1, (2010). SPIE.